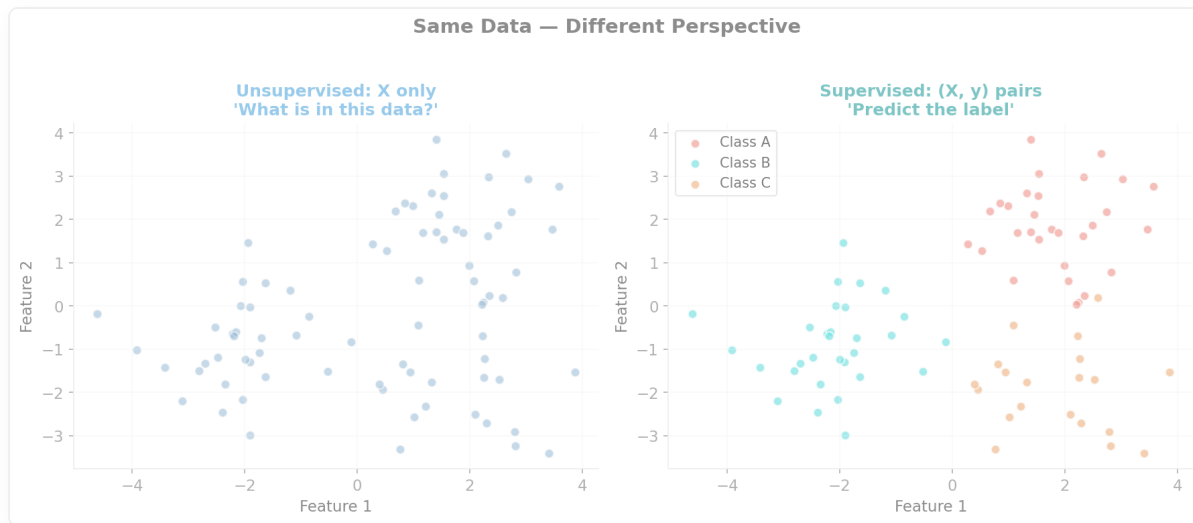


Introduction to Unsupervised Learning

Applied Machine Learning — Session 3, Chapter 1

The Key Difference



No labels. No "right answer." Just data.

Why Does Unlabeled Data Exist?

- Labeling is **expensive** (human annotation costs time + money)
- Labeling is **impossible** for future data
- Labels **don't exist yet** (discovery science)
- We don't know what we're looking for

Most data in the world is unlabeled.

Supervised learning is the exception, not the rule.

The Question Changes

Paradigm	Question
Supervised	"What is this?" (classify/predict y)
Unsupervised	"What is in this?" (find patterns)

Analogy: Sorting mail without knowing the rules.

You find your own groupings — by sender, size, topic.

Different people might sort differently. Both can be valid.

Three Types of Unsupervised Learning

1. **Clustering** (Ch08) → Group similar samples together
2. **Dimensionality Reduction** (Ch09) → Compress many features into fewer
3. **Density Estimation** → Model the probability distribution of the data

Clustering

Input: X (no labels)

Output: Cluster assignments {0, 1, 2, ...}

Before

• • ○ ○
• • • ○ ○
• • ○

→

After K-Means

[0][0] [1][1]
[0][0][0] [1][1]
[0][0] [1]

Applications:

- Customer segmentation
- Document topic modeling
- Gene expression grouping

Dimensionality Reduction

Input: X with many features

Output: X' with fewer features (preserving information)

100 features $\rightarrow$ 2 features $\rightarrow$ visualize!

Applications:

- Visualization (2D or 3D plots of high-dim data)
- Preprocessing (fewer features $\rightarrow$ faster, less overfitting)
- Noise reduction

The Evaluation Challenge

Without labels — how do we know if we did a good job?

Internal metrics (no labels needed):

- Silhouette Score: cluster separation quality
- Inertia: within-cluster compactness

External validation:

- Does it make business sense?
- Do domain experts agree?
- Does it help a downstream task?

Domain knowledge is essential in unsupervised learning.

Practical Evaluation Approach

Step 1 – Visualize: Project to 2D (PCA/t-SNE), look for structure

Step 2 – Internal metrics: Silhouette score, inertia elbow

Step 3 – External validation: Compare with known labels if available (ARI)

Step 4 – Downstream task: Does clustering improve a supervised model?

Step 5 – Domain expert: Does this grouping make real-world sense?

There is no single "correct" answer in unsupervised learning.

Multiple valid groupings can exist – choose the most useful.

Real-World Applications

Field	Application	Technique
Marketing	Customer segments	Clustering
Medicine	Disease subtypes	Clustering
NLP	Topic discovery	Clustering
Finance	Anomaly detection	Density estimation
Vision	Image compression	PCA
Any	Data visualization	t-SNE / UMAP

Let's Explore Together

→ Open `02-examples/ch07_unsupervised_intro_examples.ipynb`

We will:

1. Generate and visualize unlabeled data
2. See what "structure" looks like without labels
3. Preview what clustering will find in Chapter 8

Key Takeaways

- Unsupervised = learning without labels
- Goal: discover structure, patterns, groups
- Three main types: clustering, dim. reduction, density
- Evaluation is hard — domain knowledge matters
- Most real-world data is unlabeled → unsupervised is powerful

Next: Chapter 8

Clustering Techniques

"Let's make the machine find the groups."